LEARNER GUIDE

Area of parallelograms and trapezia

Perimeter and Area · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **parallelogram** has two pairs of parallel sides. Slide the triangle from one end to the other and it becomes a rectangle with the same base and the same perpendicular height, so its **area = base × height**. A **trapezium** has one pair of parallel sides, a and b. A copy turned half a turn fits alongside it to make a parallelogram with base a + b, so a trapezium's **area = ½ × (a + b) × height**.

What you are learning to notice

- Parallelogram: base × height. It is a rectangle in disguise.
- Sliding the top of a parallelogram sideways changes its slant, not its area.
- Trapezium: ½ × (a + b) × height, where a and b are the parallel sides.
- Always use the perpendicular height, never a slanted side.
- Every formula here comes from rearranging into a shape you already know.

Moving through it, one step at a time

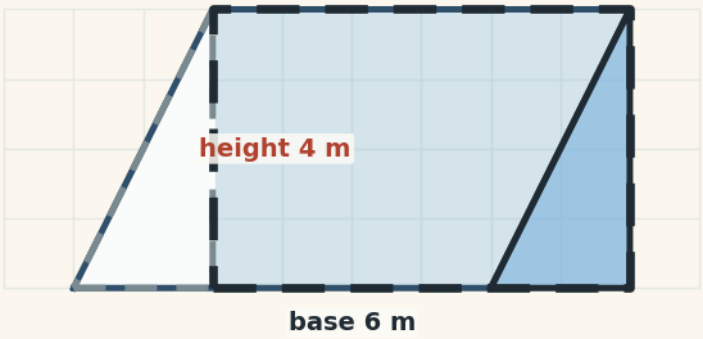
For a parallelogram, multiply the base by the perpendicular height. For a trapezium, add the two parallel sides, multiply by the perpendicular height, then halve. If you forget a formula, rearrange: slide a triangle, or add a turned copy.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A parallelogram becomes a rectangle

Find the area of a parallelogram with base 6 m and height 4 m.

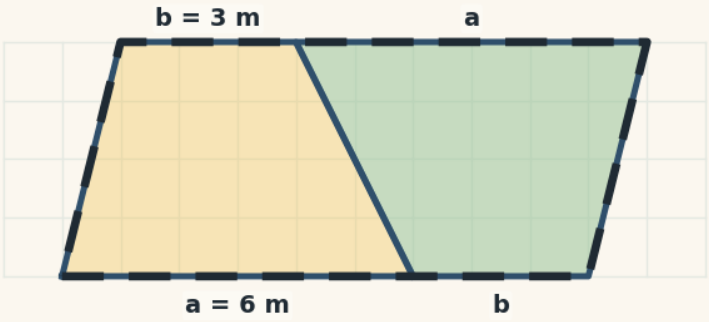


1. Cut off the triangle at one end.	a right-angled triangle
2. Slide it to the other end.	a 6 m by 4 m rectangle
3. Area of the rectangle	$6 \times 4 = 24$ m^2

Area = base × height = 24 m^2.

Example 2 — Two trapezia make a parallelogram

Find the area of a trapezium with parallel sides 6 m and 3 m, and height 4 m.



1. Add a copy turned half a turn.	a parallelogram
2. Its base is a + b.	6 + 3 = 9 m
3. The parallelogram: 9 × 4	36 m^2
4. The trapezium is half of it.	18 m^2

Area = ½ × (6 + 3) × 4 = 18 m^2.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (area of rectangles and triangles (Lesson 2)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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